W-L-122C
April 18, 1973
SUPERSEDING
Fed. Spec. W-L-122B
May 16, 1961

FEDERAL SPECIFICATION

LAMP, PHOTOFLASH

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers lamps in which a chemical reaction of combustible material sealed with a transparent bulb produces light to high intensity and short duration for photographic purposes.

1.2 Classification.

1.2.1 Classes. Lamps shall be of the following classes, and types as indicated in tables III and IV, as specified (see 6.2):

Class F. For use in essentially open-shutter photography with fast synchronized between-the-lens shutter speeds. The shape of a typical luminous flux vs. time curve is shown in figure 1A. The specified luminous flux requirement is the available total lumen-seconds between 0 and time T_2 which is the time when the shutter may close.

Class MF. For use essentially with nonadjustable front shutters, but may also cover synchronized between-the-lens shutters. The shape of a typical luminous flux vs. time curve is shown in figure 1A. The specified luminous-flux requirement is the available total lumen-seconds between 0 and T_2 which is the time when the shutter may close.

Class M. For use with conventional between-the-lens shutter synchronizers. The shape of a typical luminous flux vs. time curve is shown in figure 1B. A minimum level of luminous flux F_1 must be reached at or before time T_1 when the shutter may open. The luminous flux must then remain above this value until time T_2 when the shutter may close. However, during this period the luminous flux must not exceed F_2 in value.

Class S. For use similar to that for class F lamps except that much slower shutter speeds are used. The shape of a typical luminous flux vs. time curve is shown in figure 1A. Again, T_2 represents closing time of slowest shutter speed desirable so luminous flux beyond that time is of no use.

Class FP. For use with focal-plane shutters. The shape of a typical luminous flux vs. time curve is shown in figure 1C. The luminous flux must remain between maximum and 0.5 maximum between times T_1 and T_2 if there is to be reasonable uniformity of exposures across the negative. In addition, to avoid under-or over-exposure, the minimum luminous flux between T_1 and T_2 must exceed F_1 and the maximum must not exceed F_2. See figure 1C.

Class FC. For use with equipment that uses "Flash Cubes". The shape of a typical luminous flux vs. time curve is shown in figure 1A. The specified luminous flux requirement is the available total lumen-seconds in a zone that is a rectangle based on an angle of 20° in the horizontal and vertical between 0 and time T_2 which is the time when the shutter may close.

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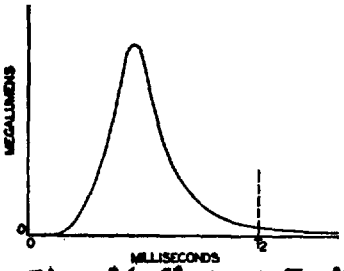


Fig. 1A-Classes F, MF, FC and S Lamps.

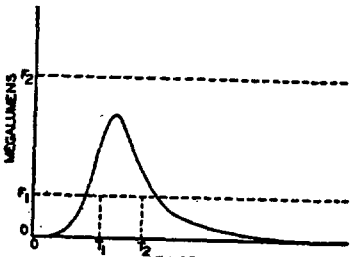


FIG.1B — GLASS M LAMPS.

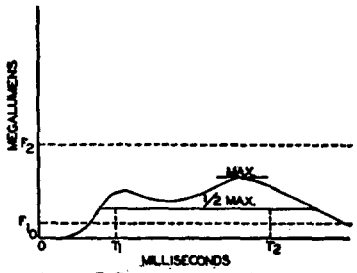


Fig.1C-Class FP Lamps.

Figure 1.-Typical luminous-flux vs. time characteristics.

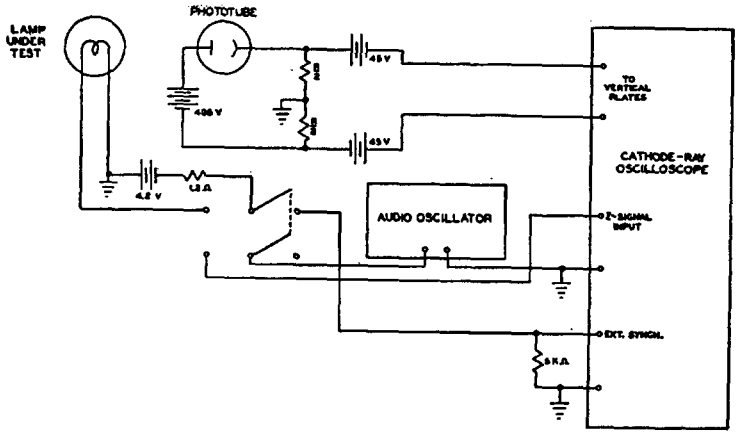


Figure 2.-Typical photometric test circuit.

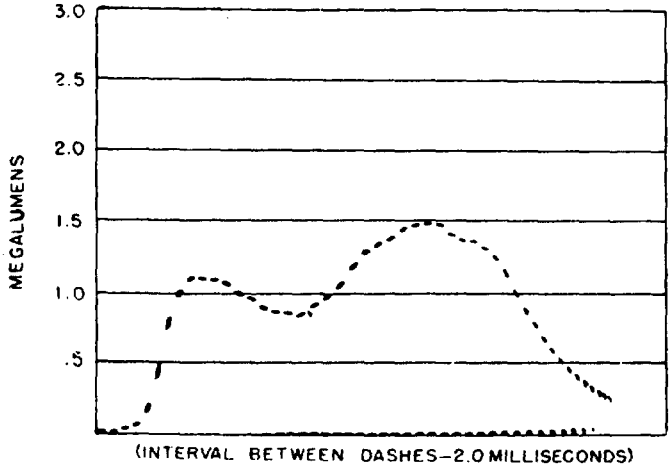


Figure 3.-Example of luminous-flux vs. time curve (class FP lamp).

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.

PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-wall.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Identification marking. The manufacturer's name, trademark, or lamp-trade identification shall be placed on the bulb, base, or disk of each lamp. The lamp-trade designation shall be placed on the bulb, base, or disk of each lamp, or on the wrapper of individual lamps or groups of lamps.

3.2 Bulb coating. Bulbs of photographic flash lamps shall be coated with a transparent lacquer which will prevent shattering of the glass.

3.2.1 Coating color. The color of the plastic coating shall be essentially clear except in the case of lamps annotated with the letter "B" in the type designation and lamps in the FC class. These lamps shall have a plastic coating, as specified in 3.2, dyed with a suitable dye to raise the color temperature of the lamp's emitted light for use with daylight type color films.

3.3 Bulb and base shapes and dimensions. Bulb and base shapes and dimensions shall be as indicated in tables III and IV and shall be in accordance with the drawings on file at the American National Standards Institute.

3.4 Operating voltage. The operating voltage range for each lamp type shall be as indicated in tables III and IV.

3.5 Luminous flux vs. time characteristics. When tested as specified in 4.2, the luminous flux vs. time characteristics of the lamps shall be in accordance with tables III and IV.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that suppliers and services conform to prescribed requirements.

4.2 Photometric tests. The photometric-test lamps shall be measured in a photoelectrically equipped integrating sphere. A luminous flux vs. time curve shall be recorded by means of a suitable oscillograph or computerized read-out system using the equipment and auxiliary circuits described in 4.2.1 or equipment that produce equivalent results. For class FC, the lamp reflector combinations shall be measured in an apparatus which measures subject illumination or horizontal beam candle power.

4.2.1 Equipment and auxiliary circuits.

4.2.1.1 Equipment shall be as follows: (1) A cathode-ray oscilloscope with Z-axis amplifier for beam modulation and a single-sweep circuit with sweep times of approximately 30 milliseconds and 100 milliseconds, (2) an oscillator capable of supplying 500 and 2,000 hertz output frequencies for connection to the Z-axis amplifier of the oscilloscope, (3) an integrating sphere of the type generally used for the measurement of the lumen output of incandescent lamps, (4) a phototube-filter combination having a spectral response approximating the luminosity curve of the International Committee on Illumination (ICI) standard observer, and (5) any camera capable of providing satisfactory photographs of the cathode-ray tube screen. The phototube circuit, connected directly without amplification to the vertical-deflecting plates of the cathode-ray tube in the oscilloscope, and the tripping circuit are shown in figure 2.

4.2.1.2 The procedure shall be as follows: The tripping switch in figure 2, normally in the right-hand position, is thrown to the left. When the right-hand contact is broken, the impulse which initiates the single sweep is applied to the external synchronisation of the oscilloscope. A few milliseconds later the left-hand contacts are closed and, simultaneously, the lamp is flashed and the oscillator output is applied to the Z-axis amplifier. The intensity adjustment of the oscilloscope is set so that the trace is faint but visible until with modulation it alternates between complete blanking out and strong intensification. An example of a luminous flux vs. time curve made with this apparatus is shown in figure 3. The zero line is obtained by operating the tripping switch without a flash lamp in the sphere.

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4.2.1.3 Calibration of the apparatus. A standard 1,500-watt incandescent lamp (approximately 30,000 lumens) is used to calibrate the Y- or lumen axis. In order to obtain deflections comparable with those obtained with flash lamps, it is necessary to replace the 2-megohm photocell load resistor, R, with a 40-megohm load resistor. This increases the deflection by the ratio of the two resistances. In addition, in order to obtain deflections of the same order of magnitude with flash lamps having widely different peak lumen outputs, a set of diaphragms of known transmission is used in the sphere window when flash lamps are measured. The calibration for the ordinates of the luminous flux vs. time curves obtained is given by the following formula:

$$\text{Lumens/inch} = \frac{F\,R_c}{D\,T\,R_t}$$

Where:

F = luminous flux of standard lamp (lumens)

D = deflection of cathode-ray beam during calibration with standard lamp operated at specified voltage and without diaphragm in sphere window (inches)

T = transmission factor of diaphragm used with test flash lamps

R_c = actual resistance of photocell load resistor used in calibration

R_t = actual resistance of the photocell load resistor used in testing, (see fig. 2)

4.2.1.4 Timing measurements are obtained from the known oscillation frequency with which the cathode beam is modulated. A frequency of 500 c.p.s. or an interval between successive dashes equal to 2 milliseconds will be found satisfactory for all lamps except class F. For class F lamps, a frequency of 2,000 c.p.s. or an interval of 0.5 millisecond between successive dashes will be found satisfactory.

4.2.1.5 The source used to flash the lamps (fig. 2) shall be 4.2 - volts in series with a resistance of 1.2 ohms. The ampere-hour capacity of the source shall be sufficiently large so that there is no appreciable change in voltage when lamps are flashed.

4.3 Acceptance inspection.

4.3.1 Inspection lot. Inspection lot shall consist of lamps of one designation and size, from an identifiable production period, from one manufacturer, and one plant, submitted for acceptance.

4.3.2 Sampling for visual, dimensional and electrical inspection. Unless otherwise specified, sampling shall be performed in accordance with the provisions set forth in MIL-STD-105 inspection level II.

4.3.3 Visual, dimensional and electrical inspection. Samples selected in accordance with 4.3.2 shall be inspected for defects as listed in table I. The AQL in percent defective is 1.5 major and 6.5 minor.

TABLE I. Defects

Major defects	Minor defects
Leaks. Short circuit. Open circuit. Unsoldered leads. Deformed bases. Cracked bulbs. Uncoated bulbs. Loose bulbs. Wrong base. Wrong bulb. Dimensions not as required.	Crooked base. Uncut lead wires. Poor lacquer coating. Wrong overall length.

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4.3.4 Sampling for photometric testing. Unless otherwise specified, sampling for photometric tests shall be performed in accordance with the provisions set forth in MIL-STD-105 inspection level II.

4.3.5 Photometric test. Lamps selected in accordance with 4.3.4 shall be tested as specified in 4.2.

4.3.5.1 Acceptance criteria for classes P, MP, PC and S. A lamp is considered defective if it gives less than the listed value of lumen-seconds during the specified time interval. The applicable AQL is 6.5 percent defective. In addition to rejection of a lot due to failure to comply with the applicable AQL, a lot may be rejected if one or more of the test lamps falls below 75 percent of the specified value for lumen-seconds in table III. Also, a lot may be rejected for reasons specified in 4.3.5.4 and 4.3.5.5.

4.3.5.2 Acceptance criteria for class M lamps. A lamp is considered defective if the instantaneous luminous flux falls outside the limits specified. The applicable AQL is 6.5 percent defective. In addition to rejection due to failure to comply with the applicable AQL, a lot may be rejected if one or more of the test lamps falls outside the specified instantaneous luminous flux for more than 25 percent of the listed time interval in table IV. Also, a lot may be rejected for reasons specified in 4.3.5.4 and 4.3.5.5.

4.3.5.3 Acceptance criteria for class FP lamps. A lamp is considered defective if: (1) It falls outside the specified limits for instantaneous luminous flux; (2) or if it falls below one-half of its observed maximum during more than 25 percent of the listed time interval in table IV. The applicable AQL is 6.5 percent defective. Also, a lot may be rejected for reasons specified in 4.3.5.4 and 4.3.5.5.

4.3.5.4 For all classes. If one of the photometric-test lamps fails to operate because of an open circuit within the lamp, it shall be excluded from the test. If more than one of the photometric-test lamps fail to operate because of an open circuit within the lamps, the lot shall be rejected.

4.3.5.5 For all classes. If one of the photometric-test lamps explodes on flashing, the occurrence shall be brought to the manufacturer's attention. The manufacturer, in the presence of the inspector, shall then take at random at leat 1 percent of the lamps remaining in the lot and flash them. At least one lamp shall be taken from each commercial shipping container. If such a supplementary test lamp explodes, the lot shall be rejected. Otherwise, the manufacturer may replace the test lamps and submit the lamps for acceptance. If more than one of the photometric-test lamps explode on firing, the lot shall be rejected without further inspection or test.

4.3.5.5.1 A lamp shall be considered to have exploded on flashing if the bulb shatters with resultant flying glass or other particles.

4.3.6 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-4 with an AQL of 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents	Number per container is more or less than required.
	Net weight exceeds requirements of the box specification.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The lamps shall be packaged in quantities as specified in the contract or order in accordance with method III of MIL-P-116.

5.1.2 Level C. The lamps shall be packaged to insure protection during shipment and safe delivery to destination.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Lamps packaged as specified in 5.1, shall be packed in quantities as specified (see 6.2) in boxes conforming to PPP-B-636, class weather-resistant or to PPP-B-640, class 2. Closure, water-proof sealing and strapping shall be in accordance with the appendix to the applicable box specification. The weight of the box and contents shall not exceed 70 pounds.

5.2.2 Level B. Lamps packaged as specified in 5.1, shall be packed in quantities as specified (see 6.2) in boxes conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636. The weight of the box and contents shall not exceed 70 pounds.

5.2.3 Level C. Lamps packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules, as applicable.

5.3 Marking. In addition to any special markings required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. All marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. All marking shall be in accordance with MIL-STD-129.

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TABLE III.

Class	Type by trade designation	Lamp flashing voltages	Physical size and shape				Nominal color temperature (^{o}K)	Luminous flux vs. time characteristic		
			Base	Bulb	Maximum diameter (inches)	Maximum length (inches)		Specification figure no.	T_2 (milliseconds after flashing)	Minimum luminous energy $0 - T_2$ (lumen-seconds)
F(Fast Peak) Lamps	SM	3 to 45	SC Bay	B-11	1.420	2-5/8	3300	1A	12	3300
	SF	3 to 45	SC Bay	B-12	1.562	2-5/8	3400	1A	12	3300
S(Slow Peak) Lamps	50	3 to 125	Med Sc	A-21	2.682	5-3/8	3800	1A	60	75,000
	3	3 to 125	Med Sc	A-23	2.938	5-15/16	3800	1A	60	90,000
MF(Medium Fast) Lamps	AG-1	3 to 45	Glass Groove	T-3-3/4	0.500	1-5/16	4000	1A	28	36,00
	M2	3 to 45	Min. No Pin	T-6-1/2	0.857	1-13/16	4000	1A	28	5400
	AG-1B	3 to 45	Glass Groove	T-3-3/4	0.500	1-5/16	5500	1A	25	2650
	M2B	3 to 45	Min. No Pin	T-6-1/2	0.857	1-13/16	5500	1A	25	2500
FC(Flash Cube)		3 to 45	Plastic	T-3-3/4		1-13/16	5500	1A	25	340 Z.L.S.*

*Z.L.S.(Zonal Lumen-seconds). The zone being a rectangle based on an angle of 20° in the horizontal and vertical.

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TABLE IV.

Class	Type by trade designation	Lamp flashing voltages	Physical size and shape				Nominal color temperature (°K)	Luminous flux vs. time characteristic				
			Base	Bulb	Maximum diameter (inches)	Maximum length (inches)		Specification figure no.	Evaluation period (milliseconds after flashing)		Instantaneous luminous flux from T_1 to T_2 (megalumens)	
									T_1	T_2	Minimum	Maximum
M(Medium Peak) Lamps	M3	3 to 45	Min. No. Pin	T-6-1/2	0.857	1-13/16	3800	1B	15	25	0.30	1.8
	5	3 to 45	SC Bay	B-11	1.420	2-5/8	3800	1B	18	26	0.25	2.0
	Press 25	3 to 45	SC Bay	B-12	1.562	2-5/8	3800	1B	18	26	0.50	2.0
	11 or Press 40	3 to 45	Med Sc	A-15	1.925	4	3800	1B	18	26	0.60	3.0
	22 or 2	3 to 125	Med Sc	A-19	2.430	5	3800	1B	18	26	1.60	5.6
	M3B	3 to 45	Min. No. Pin	T-6-1/2	0.857	1-13/16	5500	1B	15	25	0.31	0.98
	5B	3 to 45	SC Bay	B-11	1.420	2-5/8	5500	1B	18	26	0.15	1.0
FP(Focal Plane) Lamps	31 or 2A	3 to 45	Med Sc	A-21	2.682	5-3/8	3800	1C	20	60	0.30	2.5
	(FP26) (6)	3 to 45	SC Bay	(B-12 (B-11	1.562) 1.420)	2-5/8	3800	1C	15	37	0.15	0.8

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6. NOTES

6.1 Intended use. The lamps covered by this specification are intended for use with photographic equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.

(b) Specify class and type designation (see table III and IV).

(c) Levels of packaging and packing desired (see section 5).

(d) Whether civil or military marking is required (see 5.3).

(e) Quantities of lamps per container (see 5.1 and 5.2).

MILITARY CUSTODIANS:

Army - EL
Navy - AS
Air Force - 70

Review activities:

Army - EL
Navy - AS, CG
Air Force - 70

User activity:

Navy - AS

Preparing activity:

GSA-FSS

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